PROJECT 2

Exploratory Data Analysis (EDA)

DecodeLabs | Data Analytics Internship | Batch 2026

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Dataset: Sales Orders | 1,200 Rows | 2023–2025

Date: June 2026

Introduction

Project 1 answered one fundamental question: *is this data trustworthy?* By the time the data cleaning process was complete, the dataset had gone from a raw, error-prone export to a verified source of truth — 1,200 customer orders, zero duplicate IDs, properly formatted dates, and a clear separation between the 703 orders that generated real revenue ($745,088.05) and the 497 orders lost to cancellations and returns.

Project 2 asks a more strategic question: *what is this data actually telling us about the business? *

Having a clean dataset is only half the battle. The true value of analytics comes from interrogating that data — looking past surface-level totals to understand how variables interact. Are sales evenly distributed across the customer base, or are a small number of big orders skewing the average? Are certain referral channels consistently outperforming each other, or is one standout month inflating the picture?

This report walks through that analytical process. Using the cleaned dataset from Project 1, this EDA calculates core descriptive statistics, implements outlier forensics, maps feature correlations, and uncovers seasonal trends. The goal is simple: translate static rows and columns into actionable business diagnoses that a product or marketing manager can immediately use to drive strategy.

Project 1 made sure the evidence was reliable. Project 2 is where that evidence tells its story.

Methodology — The IPO Framework

This analysis follows the Input-Process-Output (IPO) framework for structured discovery:

```
[INPUT] Cleaned Sales Dataset (1,200 Rows)
    │
    ├──▶ [PROCESS] 1. Descriptive Statistics & Skewness Analysis
    │              2. Advanced Outlier Forensics (Business-Rules vs. IQR vs. Z-Score)
    │              3. Relationship, Correlation & Seasonality Mapping
    │
    └──▶ [OUTPUT] Actionable Diagnoses & Visualization Assets
```

1. Descriptive Statistics (Univariate Analysis)

I calculate the mean, median, standard deviation, and five-number summaries for TotalPrice, Quantity, and UnitPrice. Comparing the mean against the median reveals the underlying geometry of the distribution (skewness), dictating which metric truly represents a typical order.

2. Advanced Outlier Forensics

Standard analytical pipelines rely blindly on math rules. I challenge this by comparing a **Business-Rule Filter** (detecting unrealistic price floors) against two statistical methods: the **Interquartile Range (IQR) Method** and the **Z-Score Method**. This comparison highlights the difference between operational database noise and high-value business signals.

3. Bivariate Correlation & Trend Analysis

I calculate Pearson correlation coefficients (r) to mathematically define relationships between order sizes and values, analyze referral channel conversions on an Average Order Value (AOV) basis, and map monthly seasonality pivots year-over-year.

High-Level Data Profile

Before breaking down individual behaviours, this section establishes the macroeconomic baseline of the business:

Operational Metric	Value	Business Implication
Total Orders Logged	1,200	Total customer checkout volume across the platform (2023–2025).
Realized Orders (Successes)	703 (58.58%)	Orders reaching Delivered, Shipped, or Pending status.
Lost Orders (Failures)	497 (41.42%)	Transactions lost entirely to Cancelled or Returned status.
Realized Net Revenue	$745,088.05	Real money captured from successful transactions.
Lost Gross Revenue	$519,673.91	Total transactional value wiped out by failures.

A baseline failure rate of **41.42%** is exceptionally high — nearly $0.41 of every $1.00 in initiated orders is slipping through. This leakage serves as the primary lens for this exploration.

Core Insights

1. Why the Mean Overstates Our Typical Order Value

To understand customer spending patterns, I calculate descriptive statistics for TotalPrice across the full 1,200-order dataset:

Statistic	Value
Average Order Value (Mean)	$1,053.97
Middle Order Value (Median)	$823.62
Minimum Order Value	$11.39
Maximum Order Value	$3,456.40
Standard Deviation	$819.86
Skewness Coefficient	1.07 (Right-Skewed)

If we reported only the average order value of $1,053.97 to stakeholders, this would present a distorted reality. The median sits $230.35 lower at $823.62. This significant gap, paired with a positive skewness coefficient of 1.07, mathematically proves that **the sales distribution is heavily right-skewed**.

A small volume of extremely high-value transactions is pulling the mean upward, while the vast majority of everyday customers actually spend below $823. The chart below captures this visually — a heavy cluster of sales on the low-to-mid end with a long, thinning tail on the right. For business forecasting, the median is the more accurate representation of the typical customer.

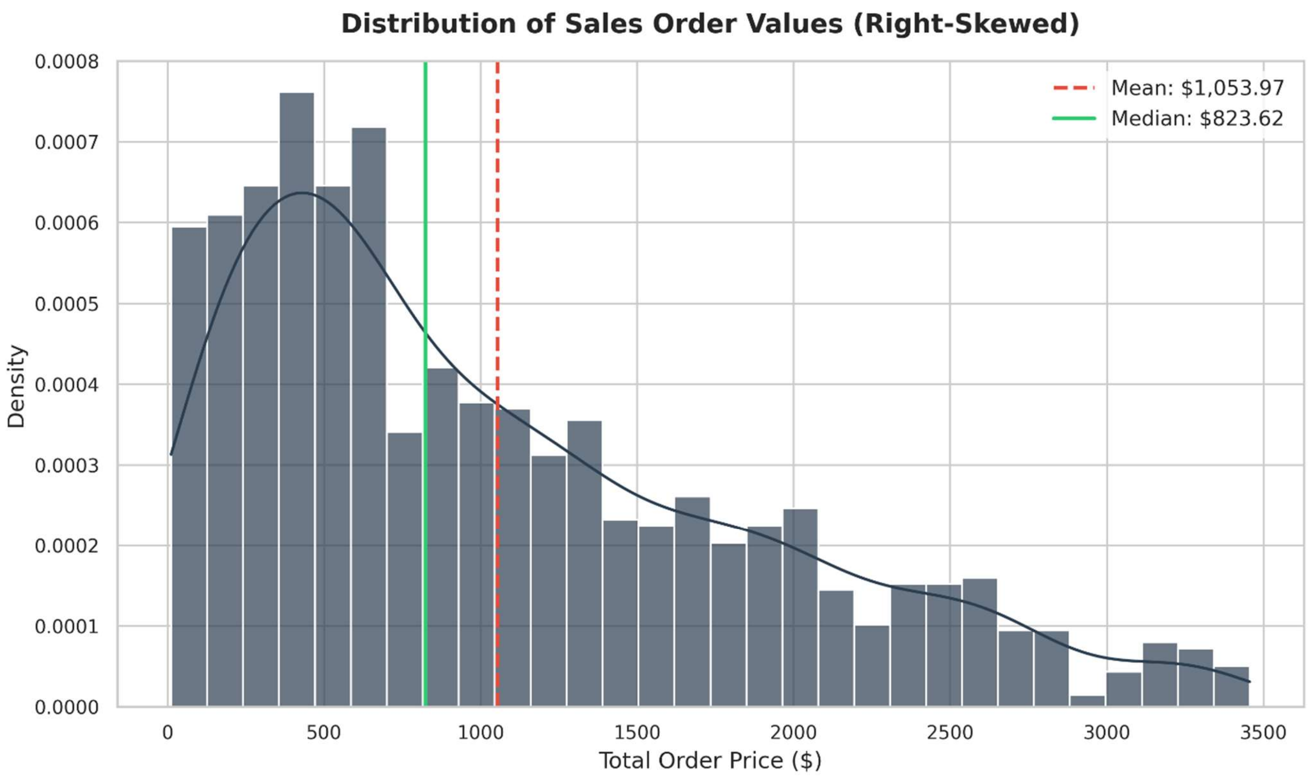

Figure 1 — Distribution of Sales Order Values showing Right-Skewed pattern with Mean ($1,053.97) vs. Median ($823.62)

2. Outlier Forensics: Business-Rule Filters Uncover Data Classification Errors

The DecodeLabs curriculum introduces standard outlier methods like IQR and Z-Score. However, blindly applying these mathematical rules can leave critical business insights completely hidden. Here, I run a comparative forensic audit using three distinct approaches:

Method A — Business-Rule Filter (UnitPrice < $50)

I apply a logical pricing floor based on product catalogue knowledge: no primary hardware product (laptops, monitors, printers, desks) should realistically sell for under $50. This filter flags **62 outlier records** across the product catalogue:

Product Category	Outlier Count	Min UnitPrice	Max UnitPrice
Printer	12	$12.10	$48.61
Monitor	11	$14.69	$49.62
Desk	11	$26.95	$46.50
Chair	9	$14.93	$46.11
Tablet	8	$17.24	$30.54
Laptop	7	$18.20	$48.38
Phone	4	$11.39	$41.90

The Diagnostic Verdict: A standard automated model would treat these low prices as natural price variations. But business logic tells us a laptop at $18.20 or a printer at $12.10 is impossible. The concentration of Printer (12) and Monitor (11) outliers points to an operational database error: **low-value accessories (ink cartridges, cables, paper) are being logged under parent hardware categories** rather than having their own SKU classifications. This noise skews unit pricing and metrics, proving that human business logic is a vital layer in data quality.

Method B — Statistical IQR Method on TotalPrice

I calculate outliers using the Interquartile Range (Q3 + 1.5 × IQR):

Metric	Value
Q1 (25th Percentile)	$410.52
Q3 (75th Percentile)	$1,578.47
IQR	$1,167.95
Lower Bound (clamped to $0)	-$1,341.41
Upper Bound	$3,330.41
Outliers Flagged	8 records (high-value bulk purchases)

The IQR method flags **8 statistical outliers** at the extreme high end (orders exceeding $3,330). These are not database errors — they represent legitimate bulk purchases such as 5 laptops or tablets ordered at once.

Method C — Statistical Z-Score Method on TotalPrice

I calculate outliers by looking for values more than 3 standard deviations from the mean ($|Z| > 3$).
Result: 0 outliers flagged.

Why did Z-Score fail? Because the distribution is heavily skewed, the extreme right-tail values pull the standard deviation up dramatically ($819.86). This inflates the Z-score boundaries, allowing extreme values to hide inside the expected range.

> *Comparative Conclusion: The IQR method is far more robust than Z-scores for real-world, skewed business datasets. But a custom Business-Rule Filter remains irreplaceable for spotting data-entry errors that no statistical method would ever catch.*

3. Feature Correlations — Defining the Drivers of Order Value

To understand what factors actually drive transactional revenue, I map the Pearson correlation coefficients (r) between the numerical attributes:

Variable Pair	Correlation (r)	Interpretation
UnitPrice vs. TotalPrice	r = 0.717 (Strong)	Premium individual items are the primary driver of large transaction totals.
Quantity vs. TotalPrice	r = 0.615 (Moderate)	Buying more items increases the total, but is not the sole driver.
ItemsInCart vs. Quantity	r = 0.650 (Moderate)	Larger carts consistently translate to higher overall quantities.
Quantity vs. UnitPrice	r = 0.015 (None)	Product pricing does not dictate bulk-purchasing behaviour at all.

The near-zero correlation between Quantity and UnitPrice (r = 0.015) is particularly insightful: customers are just as likely to buy high quantities of premium products as they are of cheap ones. This means upsell strategies based on unit price alone will not reliably increase basket size.

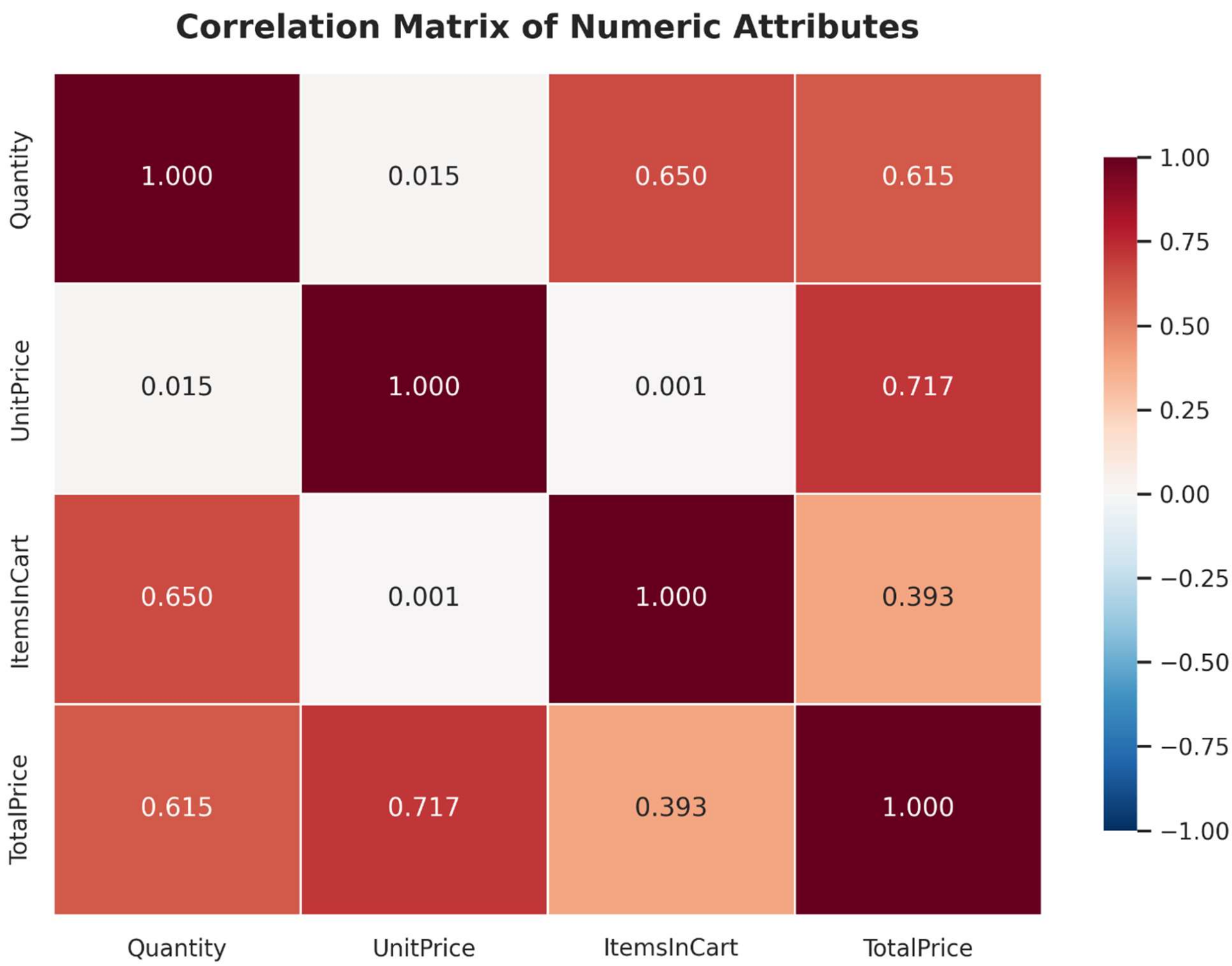

Figure 2 — Correlation Matrix of Numeric Attributes (Pearson r). Darker red indicates stronger positive correlation.

4. Seasonality — Consistent YoY Declines in January and August

A month-by-month time-series pivot reveals seasonal fluctuations. Because 2025 data only covers January through June, year-over-year trends for the second half of the year are compared across 2023 and 2024 only.

Month	2023	2024	2025	Total	Analysis
January	47	32	27	106	YoY Decline: Weakening post-holiday recovery.
February	37	32	37	106	Steady, stable volume across all years.
March	43	36	49	128	Spring rebound — strong 2025 growth.
April	31	50	32	113	Inconsistent; 2024 spike did not repeat.
May	49	34	37	120	Moderate, consistent mid-spring baseline.
June	45	53	49	147	Peak Season: Highest cumulative month.
July	44	43	—	87	Flat YoY summer plateau.
August	51	28	—	79	The August Cliff: 45% drop YoY (51→28).
September	29	44	—	73	Post-August recovery, but low baseline.
October	47	31	—	78	Autumn plateau with standard variance.
November	41	35	—	76	Standard late-year baseline.
December	46	41	—	87	Moderate holiday peak.

The June Apex: June is the high-water mark (147 total orders), driven consistently by mid-year demand across all three years.

The January Slide: January is suffering a multi-year erosion from 47 down to 27 orders — indicating a worsening post-holiday drop-off in early-year marketing momentum.

The August Cliff: In 2023, August was a top-performing month (51 orders). In 2024, it collapsed to 28. This warrants a detailed review of marketing spend or product availability during that period.

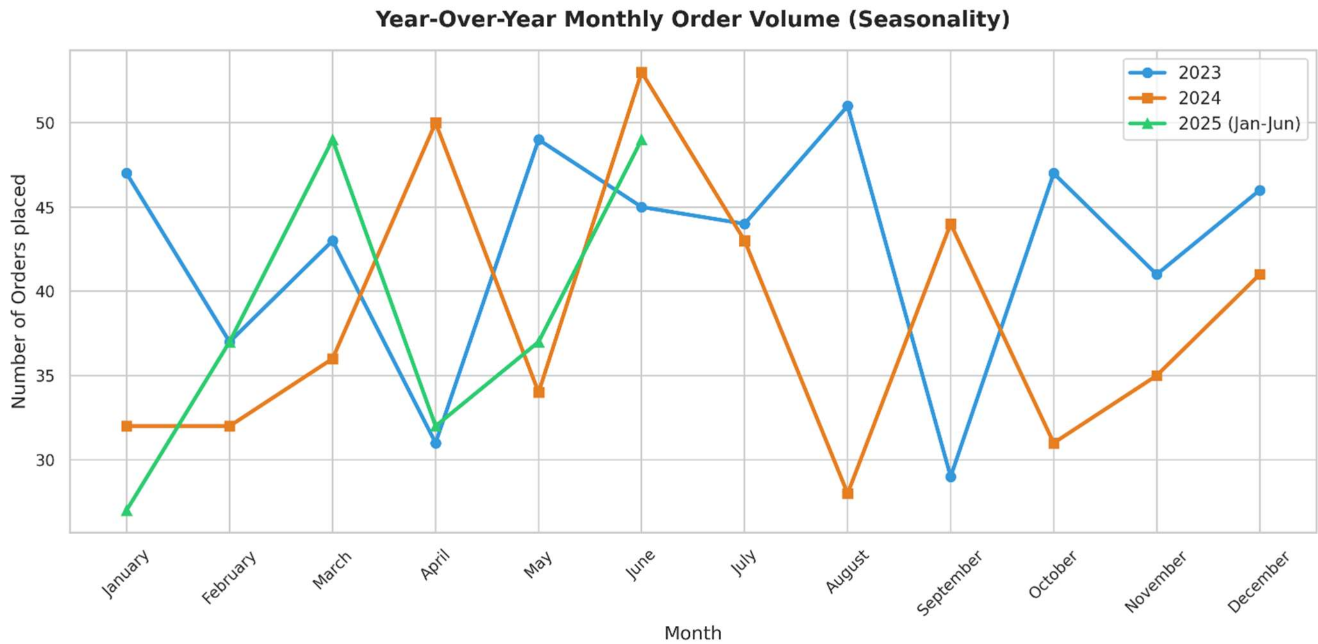

Figure 3 — Year-Over-Year Monthly Order Volume (Seasonality). 2025 covers January–June only.

5. Channel Performance — Google Leads Realized AOV, Instagram Drives Revenue

When evaluating marketing spend across the five channels, I look past simple transaction volumes. A channel bringing in hundreds of low-value, failed checkouts is a liability. Instead, I analyze **Average Order Value (AOV)**, contrasting the raw unfiltered data (all 1,200 orders) against the success-filtered data (703 realized orders).

Section A — Unfiltered View (All 1,200 Orders)

Referral Source	Unfiltered AOV	Gross Bookings	Orders Placed
1. Facebook	**$1,098.29**	$250,410.90	228
2. Instagram	$1,062.88	$275,285.45	259
3. Email	$1,047.23	$261,808.55	250
4. Google	$1,039.18	$250,441.48	241
5. Referral	$1,021.69	$226,815.58	222

Looking at all checkouts, **Facebook appears to be the premier channel** — boasting the highest AOV of $1,098.29. Instagram leads on volume with 259 orders. This is the surface-level conclusion.

Section B — Realized View (703 Successful Orders Only)

Referral Source	Realized AOV	Realized Revenue	Successful Orders	AOV Shift (Δ)
1. Google	**$1,107.75**	$155,085.42	140	**+$6.57**
2. Instagram	$1,076.72	$175,505.50	163	+$13.84
3. Email	$1,040.75	$147,787.01	142	-$6.48
4. Facebook	**$1,039.36**	$135,116.78	130	**-$58.93**
5. Referral	$1,028.07	$131,593.34	128	+$6.38

When I strip away cancellations and returns, the entire landscape changes. **Google takes the top spot** with a realized AOV of $1,107.75 (+$6.57). Instagram remains the powerhouse — generating the highest total net revenue ($175,505.50). Meanwhile, **Facebook drops from 1st to 4th place** with its realized AOV collapsing by a staggering **-$58.93 per order**.

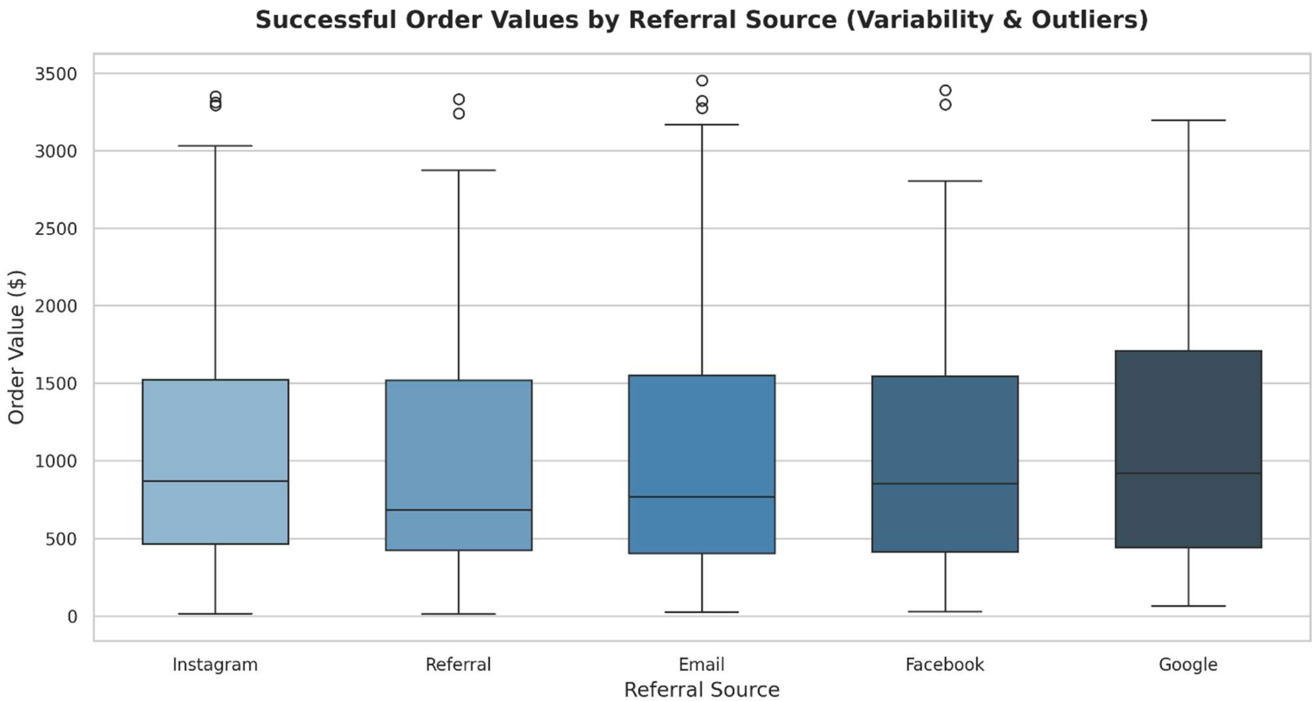

Figure 4 — Successful Order Values by Referral Source showing variability and outliers across channels.

The Facebook Targeting Paradox

This is the most valuable insight. Facebook is highly effective at driving high-intent, high-value checkouts (reflected in the unfiltered AOV of $1,098.29). However, a disproportionate share of these high-value Facebook orders are cancelled or returned before completion.

This indicates a clear **demographic or expectations mismatch** in the Facebook ad targeting:

- Our ads are reaching individuals financially capable of carting large, high-ticket hardware orders.
- However, post-checkout friction — such as extended shipping delays, mismatched product expectations from misleading ad creatives, or regional delivery issues — is causing these high-value buyers to cancel or return.
- Google and Instagram, by contrast, show **positive AOV gains** in the realized view, proving their traffic is highly qualified and highly committed.

Business Implication: Do not increase the Facebook ad budget. Instead, audit the target demographic and benchmark it against the profiles of successful Google and Instagram buyers. Review active Facebook creatives to ensure they align with actual product specifications and shipping timelines.

Actionable Recommendations

The findings from this exploratory analysis point to five clear areas where the business can take action. Each recommendation is grounded directly in what the data showed.

1. Plug the 41.4% Revenue Leakage

Nearly $520,000 in gross orders is lost to cancellations and returns. Management should initiate a secondary audit tracking specifically *why* orders fail. Are cancellations concentrated in specific regions? This points to a fulfilment or carrier issue. Are returns concentrated in specific product lines (chairs, desks)? This indicates assembly issues or inaccurate product descriptions.

2. Resolve the Product Reclassification Issue in Database Logging

The 62 low-price outliers under primary hardware categories corrupt standard product-level reporting. A dedicated Accessories product category must be introduced with separate SKUs for items like cables, paper, ink, and monitor mounts. Grouping hardware and accessories under the same parent names skews unit-price averages and misrepresents product performance.

3. Restructure Facebook Ad Targeting Parameters

Do not increase the Facebook ad budget. Audit the target demographic and adjust settings to mirror the behavioral profiles of successful Google and Instagram buyers. Additionally, review active Facebook ad creatives to ensure they align with actual product specifications and realistic shipping timelines to prevent buyer's remorse.

4. Protect and Double Down on Instagram and Google

Instagram is the highest-volume revenue generator ($175K in successful sales), and Google consistently delivers the highest-quality, committed spenders ($1,107 realized AOV). Any marketing budget freed up from inefficient Facebook campaigns should be reallocated to these two channels to maximize ROI.

5. Operational Planning for Seasonal Slumps

The business must actively plan around the January slide and the August cliff. Rather than treating these monthly drops as random occurrences, the marketing department should schedule targeted promotional campaigns specifically during these periods. Operationally, logistics and support staffing should be scaled down during these months and scaled up to support the consistent June apex.

Conclusion

Exploratory Data Analysis is the discipline of looking at data from multiple angles until the real story surfaces. As data analysts, the analyst is the translator between raw data and commercial strategy.

What the data reveals here is a business with incredible latent potential but clear operational bottlenecks. Instagram and Google are working. June is a powerhouse. The baseline net revenue of $745,088.05 is highly robust.

However, ignoring the 41% order failure rate or reporting an over-optimistic Facebook AOV based on unfiltered averages is exactly how companies burn capital. By looking closer, separating noise from signal, and applying human business logic, this analysis has identified exact areas where operational and marketing adjustments can save over half a million dollars in lost business.

The full Python code used to generate all statistics, outlier metrics, pivot tables, and visualizations in this report is available in the project repository as EDA_Python_Code_Appendix.py.

"You are the translator between data and decision." — DecodeLabs